Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Shoukas, Artin A. Ph.D.		
Project Title:	Non-Adrenergic Mechanisms of Cardiovascular Deconditioning		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ashoukas@bme.jhu.edu	**Fax:**	FY 410-614-0019
PI Organization Type:	UNIVERSITY	**Phone:**	410-955-2871
Organization Name:	The Johns Hopkins University School of Medicine		
PI Address 1:	Department of Biomedical Engineering		
PI Address 2:	720 Rutland Avenue		
PI Web Page:			
City:	Baltimore	**State:**	MD
Zip Code:	21205-2109	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	03/01/2004	**End Date:**	02/29/2008
No. of Post Docs:	3	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	4
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	15
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Berkowitz, Dan (The Johns Hopkins University) Hare, Joshua (The Johns Hopkins University) Nyhan, Daniel (The Johns Hopkins University)		
Grant/Contract No.:	NCC 9-58-CA00405		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Through our ongoing research in the area of cardiovascular de-conditioning, we have added significantly to the understanding of the potential mechanisms contributing to orthostatic intolerance in astronauts exposed to periods of microgravity. This has led to the emergence of an overarching theme that is becoming widely accepted: That orthostatic intolerance (OI) is a consequence of an adaptive neuro-humoral response in both the heart and vasculature in gravity free environment which then becomes maladaptive to the organism with rapid re-introduction of gravity. While our previous focus had remained on altered adrenergic signaling (from baroreceptor to adrenergic receptor) in the patho-physiology of orthostatic intolerance1-3, the potential role of nitric oxide, a critical modulator of both cardiac and vascular function became the focus of our investigation. In the vasculature, we have identified novel mechanisms by which NO regulates vascular endothelial function. We have also demonstrated that enhanced NO dependent signaling (as a result of NOS phosphorylation) contributes to vascular contractile hyporesponsiveness, a central component of orthostasis. We are currently using in-vivo techniques to non-invasively and serially measure vascular stiffness (as measured by pulse wave velocity) and determine the contribution of altered arterial compliance to OI. In addition, one of our graduate students, Mr. Eric Tuday, was selected as a summer intern at the Johnson Space Center, in Dr. Janice Meck's Cardiovascular Laboratories. His collaborative studies with our and Dr. Meck's group will be determining the changes in aortic and arterial compliances before, during and after long term bed rest in human volunteers. These studies test the hypothesis that long term bed rest increases arterial compliance which subsequently can decrease the cardiac output response during orthostatic challenges. With regard to the heart, we have determined for the first time that it is not loss of cardiac mass, but altered contractile function that contribute to OI. • This depressed contractile function is likely the result of altered adrenergic signaling (of which we have identified a novel arm of the beta-l beta-2 AR signaling pathway) or altered NO signaling. We have made significant contributions to the understanding of the intricate molecular mechanisms underlying the role of NO in the modulating cardiac contractile structure and function in general. • We continue to determine the role of dysregulated NO in the pathobiology of orthostatic intolerance. We now have a greater appreciation for the NO signaling dysregulation in 01 and are uniquely poised to test the efficacy of the countermeasures aimed at regulating NO as laid out in our original proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These findings could have important therapeutic significance for both astronauts and long-term bed rest patients. Both of these groups are predisposed to orthostatic intolerence upon returning to normal gravity and returning to the upright position respectively. These findings could also have important therapeutic significance for the normal aging population.
Task Progress:	1. The role of the endothelium and the NO/cGMP pathway in vascular hypo-responsiveness and remodeling in microgravity is nearly completed and will be finalized with manuscripts mid year of year 3 of this proposal. 2. The role of NO in modulating myocardial contractility and hypertrophy has just begun and we have finished the first of a series of studies during year 3, this coming year, of the project. 3. Whether up-regulation of the NO/cGMP system results in decreased vascular and ventricular stiffness as measured by integrated cardiovascular function and ventricular-vascular coupling will begin toward the end of year 3 of this project. We have delayed working on this aspect of the project since it relies on using our chronic rat preparation. This preparation requires that we purchase micro vascular pressure telemetry system to monitor the rats on a 24 hour basis.
Bibliography Type:	Description: (Last Updated: 05/05/2011)
Articles in Peer-reviewed Journals	Gregg C, Pahn A, Barouch LA, Nyhan D, Shoukas A, Hare JM, Berkowitz DE. "Beta-2 AR-coupled PI3 kinase constrains cAMP/PKA dependent increases in cardiac inotropy through cAMP dependent PDE activation." Submitted for Publication. , Jan-2006
Articles in Peer-reviewed Journals	Jhaveri R, Kim S, White AR, Burke S, Berkowitz DE, Nyhan D. "Enhanced vasodilatory responses to milrinone in catecholamine pre-contracted small pulmonary arteries." Anesth Analg. 2004 Jun;98(6):1618-22, table of contents. PMID: 15155314 , Jun-2004
Articles in Peer-reviewed Journals	Jung AS, Harrison R, Lee KH, Genut J, Nyhan D, Brooks-Asplund EM, Shoukas AA, Hare JM, Berkowitz DE. "Simulated microgravity produces attenuated baroreflex-mediated pressor, chronotropic, and inotropic responses in mice." Am J Physiol Heart Circ Physiol. 2005 Aug;289(2):H600-7. PMID: 15778286 , Aug-2005
Articles in Peer-reviewed Journals	Steppan J, Ryoo S, Schuleri KH, Gregg C, Hasan RK, White AR, Bugaj LJ, Khan M, Santhanam L, Nyhan D, Shoukas AA, Hare JM, Berkowitz DE. "Arginase modulates myocardial contractility by a nitric oxide synthase 1-dependent mechanism." Proc Natl Acad Sci U S A. 2006 Mar 21;103(12):4759-64. PMID: 16537391 , Mar-2006
Articles in Peer-reviewed Journals	White AR, Bugaj L, Attwater S, Abbot B, Li D, Champion HC, Nyhan D, Hare JM, Shoukas A, Berkowitz DE. "Early changes in vasoreactivity in a terrestrial model of microgravity are due to the upregulation of the endothelium dependent nitric oxide IcGMP pathway." Submitted for Publication. , Jan-2006
Articles in Peer-reviewed Journals	White AR, Ryoo S, Li D, Champion HC, Steppan J, Wang D, Nyhan D, Shoukas AA, Hare JM, Berkowitz DE. "Knockdown of Arginase I restores NO signaling in the vasculature of old rats." Hypertension. 2006 Feb;47(2):245-51. PMID: 16380531 , Feb-2006